



A. Color Flow Doppler (2D and 3D)

Proximal Isovelocity Surface Area

1. Align beam of flow with isovelocity beam
2. Zoomed view
3. Variance off
4. Nyquist shift
5. Measure radius (white arrow in image) from point of color aliasing to vena contracta

Vena Contracta

1. Parasternal long-axis view
2. Zoomed view
3. Best measured when proximal flow convergence, vena contracta and jet are aligned in same plane

Jet Area or Jet Area/LA Area Ratio

1. Apical view, zoom view
2. Use largest jet for measurements

Color Flow Doppler 3D

1. Use narrow color flow sector
2. Align orthogonal cropping planes
3. Planimeter vena contracta area

B. Pulsed Wave Doppler

Mitral Inflow Velocity

1. Align insonation beam with the flow
2. Sample volume at leaflet tips

Pulmonary Vein Flow Pattern

1. Use small sample volume (3-5 mm) placed 1 cm into pulmonary vein

C. Continuous Wave Doppler

Density and Contour of Regurgitant Jet

1. Align insonation beam with the flow
2. Adjust overall gain

D. Quantitative Doppler: EROA, Regurgitant Volume and Fraction

Flow Convergence Method

1. Align beam
2. Zoom
3. Look for need for angle correction (if flow convergence zone is non-planar)
4. Measure PISA radius (red solid arrow in image) at CW jet peak velocity (red dotted arrow)
5. Measure PISA radius at roughly the same time as CW jet peak velocity

Stroke Volume Method: Regurgitant Volume = SV_{LVOT} - SV_{AV}

1. LVOT systolic diameter and pulsed Doppler sample volume from different views but at same anatomic level (representing forward stroke volume)
2. Mitral mid-diastolic annulus and pulsed Doppler at the same annulus (from apical view)
3. Total LV stroke volume can also be measured by the difference between LV end-diastolic volume and end-systolic volume (best by 3D)

Advantages:

- Rapid qualitative and quantitative assessment
- This is less likely to miss patients at very large (>1.0 cm) or very small radii (<0.3 cm)

Disadvantages:

- Multiple jets, eccentric jets
- Constrained jet (LV wall)
- Non-hemiphrastic shape
- Overestimation when MRI not holystolic

Advantages:

- Surrogate for regurgitant orifice size
- Independent of flow rate and driving pressure
- Can be applied in eccentric jets
- Good at separating mild from severe MR (>0.7 cm)

Disadvantages:

- Multiple jets
- Overestimation when MRI not holystolic

Advantages:

- Multiple jets of differing directions may be measured
- Can identify severe functional MR in some cases where PISA underestimates EROA

Disadvantages:

- Subject to color Doppler blooming
- Limited temporal and spatial resolution
- Overestimation when MRI not holystolic
- Multiple jets may be in different planes, must be analyzed separately and then added

Advantages:

- E velocity >1.2 m/s a simple supportive sign of severe MR
- Diastolic A-wave inflow pattern can exclude severe MR

Disadvantages:

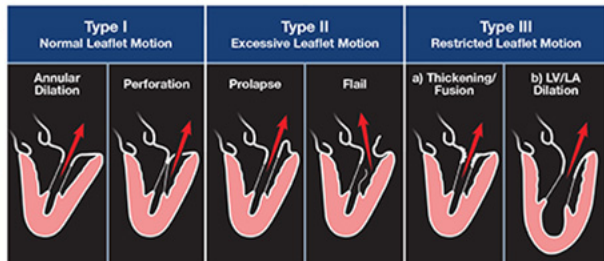
- Depends on LV relaxation and filling pressures
- High E velocity not specific for severe MR

Advantages:

- Systolic flow reversal in >1 pulmonary vein is specific for severe MR
- Normal pulmonary vein pattern suggests low LA pressure and non-severe MR

Disadvantages:

- Eccentric MR directed into a pulmonary vein may not be severe
- Systolic blurring is not specific for significant MR



Grading the Severity of Chronic MR by Echocardiography¹

Parameters	Mild	Moderate	Severe
Structural			
MV Morphology	None or mild leaflet abnormality (e.g., mild thickening, calcifications or prolapse, mild tenting)	Moderate leaflet abnormality or moderate tenting	Severe valve lesions (primary: flail leaflet, ruptured papillary muscle, severe retraction, large perforation; secondary: severe tenting, poor leaflet coaptation)
LV and LA size ²	Usually normal	Normal or mildly dilated	Dilated ³
Quantitative Doppler			
Color flow jet area ⁴	Small, central, narrow, often brief	Variable	Large central jet (>50% of LA) or eccentric wall-impinging jet of variable size
Flow convergence ⁵	Not visible, transient or small	Intermediate in size and duration	Large throughout systole
CWD jet ⁶	Faint/partial/parabolic	Dense but partial or parabolic	Holystolic/dense/triangular
VCW (cm) ⁷	<0.3	Intermediate	≥0.7 (>0.8 for bipartite) ⁸
Pulmonary vein flow ⁹	Systolic dominance (may be blunted in LV dysfunction or AF)	Normal or systolic blunting ¹⁰	Minimal to no systolic flow/systolic flow reversal
Mitral inflow ¹¹	A-wave dominant	Variable	E-wave dominant (>1.2m/sec)
Quantitative^{12,13}			
EROA, 2D PISA (cm ²)	<0.20	0.20-0.29	0.30-0.39
RVol (mL)	<30	30-44	45-59
RF (%)	<30%	30-39%	40-49%

1. Detailed signs are considered specific for their MR grade. All parameters have limitations, and an integrated approach must be used that weighs the strength of each echocardiographic measure. 2. All signs and measures should be considered in an individualized manner that accounts for age, sex, and other patient characteristics. 3. LA size and measures should be considered in an individualized manner that accounts for age, sex, and other patient characteristics. 4. Jet area is defined as the area of the jet at the narrowest point. 5. Flow convergence is defined as the area of the jet at the narrowest point. 6. CWD jet is defined as the jet area at the narrowest point. 7. VCW is defined as the width of the jet at the narrowest point. 8. Bipartite jets are defined as jets that are split into two parts. 9. Systolic flow reversal is defined as flow reversal in the pulmonary veins. 10. Systolic blunting is defined as blunting of the systolic flow in the pulmonary veins. 11. Quantitative parameters can help subclassify the moderate regurgitation group. 12. EROA is defined as the area of the jet at the narrowest point. 13. RVol is defined as the volume of the jet at the narrowest point.

